

United States Department of Agriculture
Progress Report

Title:		Climate-Smart Cotton: Developing Precision Regenerative Practices and Market Opportunities for Addressing Climate Change in the Cotton Belt	
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Non-Technical Summary

The sustainability of US cotton production is pivotal to agricultural sustainability in the Southern US. Protecting and improving soil health, reducing greenhouse gas (GHG) emissions, and enhancing input use efficiency are central to improving the sustainability of cotton in the face of changing climatic conditions. According to the most recent USDA-ERS report on tillage and conservation cropping in the US (Claassen et al. 2018), 60% of cotton acreage is still under conventional tillage, compared for example, to only 30% or less in soybean. Tillage intensity has increased due to the need to control herbicide-resistant and other problematic weeds (CAST 2012), which is now widespread throughout the US Cotton Belt (Heap 2023). Intensive tillage contributes greatly to greenhouse gas (GHG) emissions, stored carbon loss, and a loss of overall soil health, while conservation tillage is considered a vital component of a sustainable soil health management system (Elder and Lal 2008; USDA-NRCS 2017). Cotton is inherently a low residue crop, which in turn adds limited amounts of carbon during the growing season. Because cotton is commonly grown with low residue crops in rotation (Claassen et al. 2018), this is a major area where rapid gains can be achieved in soil health improvement and climate change mitigation. In this regard, cover crops and living mulches can play a key role (Wick et al. 2017). A recent analysis by the Cotton 2040 initiative revealed that cotton is a vulnerable crop to climate change impacts (WTW 2022). There is a crucial need to make cotton cultivation a carbon-neutral, or better yet a net carbon-positive enterprise, while simultaneously mitigating/adapting to the adverse effects of climate change. The majority of US cotton is already grown in marginal soils with low organic matter content. Potential increases in tillage to manage the anticipated surge in herbicide-resistant weeds associated with climate change (Ramesh et al. 2017) is expected to exacerbate

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this problem. Further, the use of Artificial Intelligence/Machine learning (AI/ML)-assisted precision application technologies can greatly improve resource input use efficiency, which can in turn reduce GHG emissions (e.g. Cotton Inc. 2018). These precision regenerative practices are also expected to create new market opportunities, through, for example, carbon crediting and sustainable cotton markets, and empower the rural agricultural workforce via employment generation.

The long-term goal of the proposed research is to apply improved precision management practices to enable enhanced carbon sequestration and reduced greenhouse gas emission, pest control, and nutrient and water management; and to address labor challenges, steward land, create new market opportunities, and ensure a sustainable supply of climate-smart cotton. Achieving this will require a coordinated effort among researchers, Extension specialists, and broader industry stakeholders. The specific objectives of this project are: 1. Develop regenerative cotton production practices and investigate their long-term effects on US Belt-wide cotton production systems. 1.1. Establish soil organic carbon and carbon intensity baselines for ecoregions in the US Cotton Belt; 1.2. Investigate regenerative practices for reducing tillage and improving soil health; 1.3. Investigate the long-term impact of regenerative practices in addressing climate change using simulation models. 2. Develop and/or evaluate precision AI/ML and smart technologies for resource conservation and climate change mitigation/adaptation in cotton. 3. Evaluate the economic feasibility of precision regenerative production practices and determine new market opportunities. 4. Expand knowledge about farmers' multi-dimensional experiences of adoption of regenerative practices across the Cotton Belt. 5. Promote the adoption of regenerative, climate-smart cotton production practices through innovative and collaborative extension and outreach activities. 6. Provide educational opportunities to train the next generation of research and extension scientists and practitioners, including underserved communities, and empower the rural workforce.

This integrated project is expected to bring tremendous benefits to the US cotton industry through the development and implementation of sound regenerative practices for reducing tillage and addressing climate-change impacts. By utilizing precision agriculture technologies, there will be an improvement in the timely diagnosis of field issues and implementation of site-specific measures for increasing input use efficiency and reducing GHG emissions. New market opportunities will be generated, improving overall farm profitability. The economic analysis will highlight the economic benefits of the adoption of precision regenerative practices and smart-technologies for cotton production. The health and safety of farm workers will be improved. Automation will help alleviate labor shortage in rural areas, while equipping them for new employment opportunities.

Accomplishments

Major goals of the project

The **long-term goal** of the proposed research is to apply improved precision management practices to enable enhanced carbon sequestration and reduced greenhouse gas emission, pest control, and nutrient and water management; and to address labor challenges, steward land, create new market opportunities, and ensure a sustainable supply of climate-smart cotton.

The specific objectives of this project are:

1. Develop regenerative cotton production practices and investigate their long-term effects on US Belt-wide cotton production systems
 - 1.1. Establish soil organic carbon and carbon intensity baselines for ecoregions in the US Cotton Belt
 - 1.2. Investigate regenerative practices for reducing tillage and improving soil health
 - 1.3. Investigate the long-term impact of regenerative practices in addressing climate change using simulation models
2. Develop and/or evaluate precision AI/ML and smart technologies for resource conservation and climate change mitigation/adaptation in cotton
3. Evaluate the economic feasibility of precision regenerative production practices and determine new market opportunities
4. Expand knowledge about farmers' multi-dimensional experiences of adoption of regenerative practices across the Cotton Belt
5. Promote the adoption of regenerative, climate-smart cotton production practices through innovative and collaborative extension and outreach activities
6. Provide educational opportunities to train the next generation of research and extension scientists and practitioners, including underserved communities, and empower the rural workforce

What was accomplished under these goals?

During the reporting period, the project made strong progress across multi-state field research, soil and microbial analyses, regenerative practice evaluation, precision technology development, economic analysis, and farmer adoption studies. Activities were implemented across Texas, New Mexico, Arizona, Georgia, North Carolina, Mississippi, Alabama, and Tennessee, generating important datasets and strengthening the foundation for future research, outreach, and training.

Objective 1. Develop regenerative production practices and determine their long-term effects on US cotton.

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During this reporting period, the project team made strong progress in evaluating regenerative cotton production practices across multiple states. In Texas, Dr. Jake Mowrer's team collected soil samples from cotton sites under five tillage and cover crop treatments. Soil organic carbon, respiration, and wet aggregate stability measurements were completed, with AMF and phosphatase analyses underway. Dr. Katie Lewis completed the first cotton/forage sorghum rotation cycle and assisted with soil sampling from 20-30 High Plains farms. Dr. Nithya Rajan's team evaluated bed configuration and cover crop termination timing in Texas Blackland Vertisols, with preliminary results showing that cover crops maintained or improved yield performance, especially under the two-week termination treatment.

In New Mexico, Dr. Rajan Ghimire's team completed first-year soil health data collection and analysis for the circular grass buffer strips project, collected soil profile carbon and nitrogen samples, identified on-farm sampling sites, presented preliminary results, and had one manuscript accepted. In Arizona, Dr. Debankur Sanyal demonstrated cover crop production on beds in rotation with upland cotton under irrigated conditions, with cover crop treatments producing higher lint yield than controls. In Georgia, Dr. Nicholas Basinger completed SOC and bulk density sampling from 20 fields and AMF sampling from 10 fields. In North Carolina, Dr. Ramon Leon-Gonzalez collected soil and mid-season AMF samples from five cotton fields, with additional sampling planned from 25 fields. Microbiome and root-trait work also advanced. Dr. Sanjay Antony-Babu's team completed DNA extraction and 16S sequencing for samples from Brazos Bottom and Lamesa, showing distinct microbial signatures between different production systems. Auburn teams evaluated cotton lines and cultivars for weed suppression, photosynthetic traits, and root traits. Mississippi State completed all 30 on-farm soil sampling sites and collected key seasonal crop, soil, and cover crop measurements. Agricenter International implemented field treatments, including flame weeding, mulch, cultivation, and autonomous tractor operations, with data collected on labor, weed control, plant response, and yield.

Objective 2. Develop and/or evaluate precision AI/ML and smart technologies for resource conservation and climate change adaptation in cotton.

Progress focused on building AI/ML-ready datasets and advancing simulation tools for climate-smart cotton production. Dr. Mahendra Bhandari and Dr. Uday Vaddevolu recruited Sabin Shrestha, who began his Ph.D. program in Spring 2026. The team also developed a comprehensive data acquisition plan with research programs, institutes, and industry partners involved in irrigation research. This plan will integrate data from soil moisture sensors, UAS platforms, satellite imagery, and agronomic measurements, including plant height, biomass, yield, fiber quality, irrigation amounts, and treatment strategies. These multi-source datasets will help assess existing data resources and identify additional experimental needs for training and validating AI/ML models across cotton-growing regions.

GaiaDhi AgTech, in collaboration with Texas A&M, advanced simulation-based evaluation of regenerative practices for climate change adaptation. The Holos DSS model was configured and calibrated for the TAMU cotton test farm, with validation testing completed for yield. The team extracted modeled soil CO₂ emission trends across farming practices, simulated planting date effects on cotton yield, and initiated crop rotation simulations to evaluate climate-friendly and adaptive cotton production strategies.

Objective 3. Evaluate the economic feasibility of precision regenerative production practices and determine new market opportunities.

Progress focused on evaluating the economic feasibility, profitability, risk, and market potential of precision regenerative cotton practices. Dr. Darren Hudson's team completed a meta-analysis of U.S.-based agronomic studies on precision application and soil health practices, including guidance systems, variable rate application, cover crops, and pasture management. For cotton, guidance systems and variable rate planters were identified as important practices for improving yield and profitability. The team also analyzed risk and insurance impacts of soil health improvements using Monte Carlo simulation and Texas High Plains data. Results showed that improved soil health can increase yield and reduce crop failure risk in dryland cotton, with yield benefits outweighing higher insurance costs.

Objective 4. Expand knowledge about farmers' multi-dimensional experiences of adoption of regenerative practices across the Cotton Belt.

Progress focused on understanding producer adoption of regenerative practices and the social dimensions of soil health management. Dr. Jennifer Jo Thompson's team facilitated three in-person focus groups in South Georgia with nine current or prospective cotton farmers, in collaboration with the Soil Health Institute, to examine adoption challenges related to cover crops and other regenerative practices. Results were analyzed and shared through an internal report, social media post, UGA Impact Statement, and a presentation at the 2025 American Anthropological Association meeting, with a manuscript in progress. The team also supported learning through the 2026 Southern Cover Crop Council Conference, farm visits, and farmer discussions. A field survey instrument was developed with UGA and SHI collaborators for Georgia producers.

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participating in SAS CAP soil sampling, with plans to share it across other Cotton Belt teams after piloting.

Objective 5. Promote the adoption of regenerative, climate-smart cotton production practices through innovative and collaborative extension and outreach activities.

Extension and outreach activities are planned for the later project years, with Dr. Benjamin McKnight's component scheduled for 2028-2029. Dr. Peter Dotray shared SAS Smart Cotton research updates through county meetings, a public radio segment, agricultural conferences, and university symposiums, reaching farmers, consultants, industry, students, and stakeholders. Four undergraduate students also gained hands-on training in field and greenhouse weed science activities.

Objective 6. Provide educational opportunities to train the next generation of research and extension scientists and practitioners, including underserved communities, and empower the rural workforce.

The project also contributed to student training and workforce development. Four undergraduate students from Texas Tech University were employed and supported through the project and gained hands-on experience in field and greenhouse research related to chemical and non-chemical weed control tactics in agronomic cropping systems. These students were trained in weed identification, sprayer calibration, small plot data collection, data entry, and exposure to various farming practices. More broadly, project activities across locations continued to provide practical research experience and professional development opportunities for students and trainees involved in regenerative and climate-smart cotton research.

What opportunities for training and professional development has the project provided?

During the 2025 to 2026 reporting period, the project provided meaningful opportunities for training and professional development for undergraduate and graduate students, postdoctoral researchers, and early-career professionals across participating institutions. These activities supported the project goals by strengthening technical skills in regenerative cotton production, soil health assessment, weed management, microbiome analysis, precision agriculture, and stakeholder engagement, while also helping prepare the next generation of research and extension professionals.

At Texas A&M University and collaborating locations, students and trainees were engaged in field, laboratory, and data-driven research activities. Project-related work provided hands-on experience in soil sampling, biomass collection, treatment implementation, yield sampling, and assessment of soil biological and chemical indicators, including soil organic carbon, respiration, AMF, phosphatase activity, and related soil health measurements. Additional professional development was supported through participation in microbiome and hyphal recruitment studies, where trainees gained experience in DNA extraction, sequencing workflows, community analysis, and interpretation of biological responses under different cotton management systems. In the precision agriculture area, recruitment of graduate students and the development of data integration frameworks created opportunities for training in data compilation, preprocessing, feature engineering, model development, and use of multi-source datasets involving soil moisture sensors, UAS, satellite imagery, and agronomic measurements.

The project also supported direct student training through field and greenhouse activities. Four undergraduate students from Texas Tech University were employed and supported through the project and received hands-on training in chemical and non-chemical weed control tactics used in agronomic cropping systems. Their training included weed identification, sprayer calibration, small plot data collection, data entry, and exposure various farming practices. Additional educational opportunities will continue through undergraduate student employment and participation in project activities in Texas and other locations.

Professional development was further strengthened through classroom instruction, workshops, field days, and scientific presentations. At Texas A&M University, project-related content was shared through **SCSC 301 Soil Science**, and **PSS 4421 - Principles of Weed Science** will continue to provide training opportunities linked to the project. At Auburn University, instructional efforts included the development of a module on the importance of roots for crop growth, including discussion of water and nutrient uptake and approaches for studying roots in the laboratory and field. Across institutions, project personnel and trainees also benefited from presenting or participating in professional meetings and symposia, including **CANVAS 2025**, the **Beltwide Cotton Conferences**, the **Texas A&M Postdoctoral Research Symposium**, the **Graduate Student Research Symposium**, **STEGG 2025**, and the **American Phytopathological Society Annual Meeting**. These activities expanded scientific communication skills, increased exposure to interdisciplinary research communities, and supported networking and collaboration.

In addition to technical and academic training, the project created opportunities for professional growth through engagement with growers, extension personnel, and other stakeholders. Field days, grower meetings, workshops, and extension tours across Texas, Georgia, Mississippi, Arizona, and other project regions exposed students and early-career professionals to applied challenges and stakeholder perspectives in climate-smart cotton production. These experiences helped connect research activities with real-world production systems and strengthened participants' understanding of how research,

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extension, and producer needs interact across the Cotton Belt.

Overall, the project has provided a strong platform for experiential learning and professional development by combining field research, laboratory analysis, classroom instruction, scientific communication, and stakeholder engagement. These activities have helped build technical capacity, research experience, and professional skills among students and trainees while supporting the broader goals of regenerative and climate-smart cotton research.

How have the results been disseminated to communities of interest?

Project results were disseminated through a combination of field days, grower meetings, producer workshops, conference presentations, classroom teaching, webinars, technical seminars, stakeholder visits, and social media/outreach products. These activities reached a broad audience, including farmers/producers, consultants, industry representatives, students, faculty, Extension personnel, government and agency partners, local stakeholders, pest control advisors, USDA/NRCS professionals, and research collaborators.

Research and outreach updates were shared through formal academic and professional venues. Dr. Jake Mowrer disseminated project information through **SCSC 301 Soil Science** at Texas A&M University and through participation in **CANVAS 2025** in Salt Lake City, reaching students and industry audiences. Dr. Sanjay Antony-Babu's team presented project findings at several regional, national, and institutional scientific meetings, including the **2026 Beltwide Cotton Conferences, CANVAS 2025, Texas A&M Postdoctoral Research Symposium, Graduate Student Research Symposium, Southeast Texas Evolutionary Genetics & Genomics Symposium**, and the **American Phytopathological Society Annual Meeting**. These presentations helped share updates on microbiome-driven soil health strategies and supported collaboration across microbiology, plant pathology, and agronomic research communities.

Several field-based outreach events directly engaged producers and agricultural stakeholders. Dr. Katie Lewis hosted a field day with nearly **100 attendees**, including approximately **15 farmers, 20 industry/government representatives, 35 students**, and **30 faculty**, where cotton cropping systems under the SAS project were discussed. Dr. Nicholas Basinger shared project information at the **J. Phil Campbell REC Cotton Field Day** in Watkinsville, Georgia, reaching about **80 participants** through the cotton field day and corn boil. Dr. Debankur Sanyal demonstrated project research at the **Central Arizona Field Day** and provided cotton research updates through four **2025 cotton tent talks** on May 27, July 16, August 5, and September 9, reaching about **50 participants**, including growers, consultants, industry representatives, local stakeholders, pest control advisors, USDA researchers, NRCS professionals, and others.

In New Mexico and surrounding regions, Dr. Rajan Ghimire's team shared results on soil health, organic carbon, and circular buffer strips through a **Royal College UK webinar** with an estimated **100 participants**, a special seminar at Colorado State University, and multiple presentations to farmers in eastern Colorado. Dr. Darren Hudson shared economic and risk management information through the **Olton Alternative Crops Conference** in Texas, targeting farmers and producers. Dr. Jennifer Jo Thompson's team conducted **three creative workshops** in Georgia focused on collective knowledge around soil health, reaching **30 farmers/producers, 1 Extension officer**, and **2 educators**.

Mississippi State University also conducted extensive dissemination through field days, grower meetings, producer meetings, and irrigation-related presentations. Dr. Drew Gholson's team reported a field day with **45 participants**, a grower meeting with **76 participants**, and producer meetings with **61 participants** across multiple Mississippi counties, including Prentiss, Hinds, Humphreys, Leflore, Holmes, Yazoo, Tunica, Grenada, Noxubee, and Monroe counties. Additional presentations were delivered through the Governor's Water Task Force, Mississippi Master Irrigator Program, Water for Food Global Conference, Mississippi Agricultural Consultants Association Conference, MSU Row Crop Short Course, Delta Ag Expo, Delta Area Rice Meeting, and MAIC Row Crop Certified Crop Advisors Program.

In Texas, Dr. Peter Dotray shared SAS Smart Cotton research updates through the **Sandy Land Crops Conference** in Seminole, Texas, reaching approximately **250 participants**, and the **Resilient Cropping Systems Tour** in Plainview, Texas, reaching approximately **100 participants**. These events engaged farmers, consultants, industry representatives, students, and local stakeholders. At Agricenter International, Andrew Jordan and Ashley Barth supported dissemination through **five Ag Launch robotics collaborator visits** in July and August 2025 and field visits with U.S. Cotton Trust Protocol staff in October 2025, reaching consultants, industry partners, and local stakeholders.

Overall, events with reported participant numbers reached at least **904 individuals**, in addition to several academic conferences, classroom activities, seminars, webinars, public presentations, social media posts, stakeholder visits, and professional meetings where attendance numbers were not fully reported. Together, these dissemination activities helped share project outcomes with scientific, producer, student, industry, and Extension audiences across the U.S. Cotton Belt.

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What do you plan to do during the next reporting period to accomplish the goals?

During the next reporting period, the project team will continue field implementation, laboratory analyses, data integration, modeling, outreach, and student training activities to accomplish the goals of the SAS Cotton project. Planned work will build on the current year's progress across regenerative production practices, soil health, AI/ML-based decision support, economic assessment, producer adoption, outreach, and workforce development.

For soil health and regenerative production systems, Dr. Jake Mowrer's team will conduct additional soil respiration measurements to obtain third-year data for evaluating CO₂ emission trends. Field sampling will continue across all locations to strengthen temporal comparisons of carbon dynamics. Laboratory work will focus on AMF colonization and phosphatase enzyme activity to better understand biological responses to conservation management. The integrated dataset will be used to evaluate relationships among soil biological activity, carbon cycling, and long-term soil health improvement. Dr. Katie Lewis's team will continue the cotton/forage systems into year 3 with the addition of nitrogen fertility treatments to evaluate interactions among nitrogen rate, cover crop species, crop rotation, and nitrogen cycling. Additional soil chemical, biological, and physical measurements will be collected to assess impacts on nutrient availability and system performance. Dr. Rajan Ghimire's team will continue soil health assessment in the circular grass buffer project, conduct on-farm soil sampling and analysis, publish at least one research article from soil health data, and present results at professional and Extension/outreach events.

Field-based regenerative cotton studies will also continue across multiple states. Dr. Nicholas Basinger's team will continue data collection from the previous season, add biomass measurements, plant cotton into living mulch, continue sampling 25 fields, and conduct herbicide screening in both greenhouse and field settings. Dr. Debankur Sanyal's team has already planted cover crops in Arizona and will plant the second season of cotton in late April and early May. Soil samples collected at cover crop planting will be followed by two additional soil sampling events, and cover crop biomass will also be collected. Dr. Ramon Leon-Gonzalez's team will complete sampling of the remaining 25 fields in North Carolina, with Extension agents and growers contacted to identify suitable fields and soil core sampling equipment to be rented to speed up collection. The team will also collect the second year of CE2 data following the shared protocol. Dr. Nithya Rajan's team will continue field implementation at Texas sites, including treatment maintenance, cover crop management, planting, field monitoring, biomass and yield sampling, and other measurements required by project protocols.

Research on crop-weed interactions, microbiomes, and root traits will be advanced through field, greenhouse, and laboratory studies. Drs. Alvaro Sanz-Saez's and Nithya Subramanian's teams will repeat the cotton germplasm field experiments and continue cotton root imaging and characterization. Dr. Sanjay Antony-Babu's team will expand temporal and management-driven analyses of hyphal recruitment dynamics in the Brazos Bottom fields through additional seasonal sampling, completion of alpha and beta diversity analyses, functional prediction analyses, and integration of hyphosphere recruitment data with soil community datasets. For the cotton microbiome study, the team will complete remaining leaf DNA extractions, conduct 16S rRNA metabarcoding for leaf samples, analyze soil and leaf microbiome community composition, conduct functional prediction and network analyses, and integrate soil and leaf datasets to assess cross-compartment microbial dynamics. Dr. Alvaro Sanz-Saez will record two teaching sessions on the importance of roots for crop growth, root water and nutrient uptake, and approaches for studying roots in the lab and field, making the module available for the project team and future classes.

For precision technologies, AI/ML, and climate adaptation, Dr. Mahendra Bhandari and Dr. Uday Vaddevolu's team will focus on developing the data integration framework and initial model development for precision irrigation systems. They will formalize data-sharing collaborations with research programs, institutes, and industry partners and compile multi-source datasets across cotton-growing regions. Data harmonization, standardization, and quality control protocols will be implemented to create a structured and scalable database. Targeted field experiments will be designed and initiated, if needed, to address data gaps. The Ph.D. student will lead data compilation, preprocessing, feature engineering, and preliminary model development to evaluate irrigation response, crop growth, and yield outcomes. Prototype models will be developed and validated using historical and newly collected datasets. Dr. Drew Gholson's team will recruit an MS student to support Objective 2 work on irrigation scheduling advances using AI/ML and integrating historical and current sensor data to reduce the cost of scheduling tools for cotton growers.

Climate-smart modeling and economic assessment will continue to support decision-making and adoption. GaiaDhi AgTech, in collaboration with Texas A&M, will develop a paper on planting date impacts, continue and productize crop rotation modeling for project use, expand solutions to larger regions of the Cotton Belt, and investigate proximal sensing integration into models for in-season practice adjustments. Dr. Darren Hudson's team will complete economic and risk analyses and pursue publication of findings in academic outlets. As additional primary data are collected by project collaborators, the team will begin economic analysis of experimental results to evaluate profitability, risk, and adoption implications of regenerative

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and precision practices.

Producer adoption and social science activities will continue in Georgia and across the Cotton Belt. Dr. Jennifer Jo Thompson's team will continue field survey data collection in Georgia and share the survey instrument with SAS CAP teams in other Cotton Belt regions for producer implementation. The team will facilitate additional participatory workshops in collaboration with the Soil Health Institute to investigate producers' perspectives on soil health and soil health reporting. Data from surveys and workshops will be analyzed, reports will be prepared to summarize methods, results, and conclusions, and one manuscript will be submitted for peer review.

Outreach, Extension, and workforce development activities will also continue. Dr. Peter Dotray's team will present research and Extension updates at the Organic Cotton and Peanut Tour in Seminole, Texas, during summer 2026, along with other county Extension meetings in spring 2026. The course PSS 4421 - Principles of Weed Science will be taught again in fall 2026. The team will also employ spring and summer undergraduate student workers to gain hands-on training in chemical and non-chemical weed control tactics used in Southern High Plains cropping systems. At Agricenter International, Andrew Jordan and Ashley Barth will continue experimental design, field preparation, land preparation, fertilizer planning, and resource acquisition for upcoming field activities.

Participants

Actual FTE's for this Reporting Period

Role	Non-Students or faculty	Students with Staffing Roles			Computed Total by Role
		Undergraduate	Graduate	Post-Doctorate	
Scientist	4	0	0	0	4
Professional	0.1	10.5	10.4	2.4	23.4
Technical	3.3	0	0	0	3.3
Administrative	1.3	0	0	0	1.3
Other	0	0	0	0	0
Computed Total	8.7	10.5	10.4	2.4	32.0

Student Count by Classification of Instructional Programs (CIP) Code

Undergraduate	Graduate	Post-Doctorate	CIP Code
5	4	2	01.00 Agriculture, General.
5	4	2	01.11 Plant Sciences.
4	3	1	01.12 Soil Sciences.

Target Audience

The key target audiences for this project include cotton farmers across Texas and the broader U.S. Cotton Belt, with particular emphasis on those who are interested in adopting, evaluating, or expanding precision regenerative and climate-smart production practices. This includes producers working under conventional, conservation, and regenerative systems, as well as those seeking practical approaches to improve soil health, input efficiency, water management, crop resilience, and long-term sustainability under changing climatic conditions.

Another major audience includes agricultural professionals who directly support producer decision-making, such as county and regional extension agents, crop consultants, certified crop advisers, pest management advisers, and other technical service providers. These groups play an essential role in translating project findings into practical recommendations for cotton production systems across different regions. Their engagement is important for expanding the reach of project outcomes and supporting the adoption of tested practices and decision-support tools.

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The project also focuses on research, education, and institutional audiences. These include undergraduate and graduate students, postdoctoral researchers, and early-career professionals involved in field research, laboratory analyses, precision agriculture, soil health assessment, agronomy, weed science, and related disciplines. By involving these groups in hands-on training, field experiments, data analysis, and outreach activities, the project helps build the next generation of research and extension professionals prepared to work in climate-smart and regenerative cotton systems.

In addition, the project is relevant to a broader network of stakeholders connected to cotton production and sustainability efforts. These include agricultural industry representatives, input and technology providers, irrigation and water management professionals, ginners, merchandisers, supply chain partners, and organizations interested in sustainable cotton production and market opportunities. These audiences are important because project outcomes are intended not only to improve field-level management but also to support broader system-level advances in profitability, efficiency, sustainability, and resilience across the cotton sector.

The project also serves researchers, professional societies, collaborating institutions, and public and private organizations involved in soil health, climate adaptation, carbon-related outcomes, and agricultural innovation. This includes stakeholders interested in the economic, environmental, and social dimensions of regenerative practice adoption, as well as those working on data-driven technologies such as AI/ML, remote sensing, digital agriculture, and irrigation decision-support systems. These audiences will benefit from the project's multi-state datasets, applied research findings, and integrated analyses across agronomic, economic, and social science components. A further important audience includes rural communities and underserved groups who may benefit from improved access to educational opportunities, training, and workforce development supported by the project. Through student engagement, experiential learning, outreach events, and collaborative activities, the project aims to strengthen workforce capacity and create broader awareness of climate-smart cotton production systems.

In addition to these primary audiences, broader stakeholders and members of the public interested in sustainable agriculture, regenerative practices, and climate-smart farming will also be served through field days, workshops, grower meetings, scientific presentations, educational materials, and other outreach efforts. By engaging a diverse range of stakeholders across production, education, research, technology, and industry sectors, the project seeks to create a strong network of knowledge exchange and practical application. Collectively, these efforts will help ensure that project outcomes are translated into actionable strategies, support the adoption of precision regenerative practices, and strengthen the long-term sustainability and resilience of cotton production across the U.S. Cotton Belt.

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Products

DR 1020-006 Certification Statement

Yes, any applicable scholarly publications have been submitted to the National Agriculture Library's (NAL) PubAg and any applicable data assets have been catalogued in NAL's Ag Data Commons. These entries included the NIFA grant or accession number and digital persistent identifiers for the publication of data asset (such as DOI) and the authors (such as ORCID).

Type

Conference Papers and Presentations

Status

Published

Year Published

2025

NIFA Support Acknowledged

YES

Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Ramachandran, R., Mowrer, J. E., Bagavathiannan, M., Rajan, N., & Isakeit, T. (2025). 120 years of cotton cultivar development: Impacts on root morphology and potassium response [Abstract]. Texas Plant Protection Association 2025, Bryan, TX.

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Citation

Ghimire, R., Sapkota, S., Frene, J. P., Salehin, S. M. U., & Angadi, S. V. (2025, April 9). Soil health and organic carbon responses to the circular buffer strips of perennial grasses [Webinar]. Royal College, UK.

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Citation

Ghimire, R., & Angadi, S. V. (2025, April 2). Circular buffer strips of native perennial grasses: An innovative way to add multiple benefits to center pivot irrigated agriculture [Special seminar]. Colorado State University, Department of Soil and Crop Science, Fort Collins, CO.

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2025

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YES

Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Ghimire, R., & Angadi, S. V. (2025, March 31–April 1). Soil health and organic carbon responses to the circular buffer strips of perennial grasses [Farmer presentations]. Eastern Colorado.

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Conference Papers and Presentations

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Author ORCID(s)

Citation

Mollaee, M., Coello, S. G., Sanz-Saez, A., Bagavathiannan, M., & Maity, A. (2026, March 8–10). Field evaluation for allelopathic potential of 120 cotton lines against southern weeds for sustainable weed management. In Proceedings of the Southern Weed Science Society Annual Meeting, Nashville, TN.

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Author ORCID(s)

Citation

Nimakoh, M., Hudson, D., & Bagavathiannan, M. (2026). An analysis of precision management technologies and agricultural productivity in the U.S. [Presentation]. Southern Agricultural Economics Association Annual Meeting, Louisville, KY.

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Author ORCID(s)

Citation

Taylor, C., Hudson, D., Che, Y., & Wang, C. (2026). Potential insurance implications of climate-smart production practices [Presentation]. Southern Agricultural Economics Association Annual Meeting, Louisville, KY.

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Citation

Nimakoh, M., Hudson, D., & Bagavathiannan, M. (2026). An analysis of precision management technologies and agricultural productivity in the U.S. [Presentation]. Beltwide Cotton Conferences Economics and Marketing Conference, San Antonio, TX.

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Citation

Taylor, C., Hudson, D., Che, Y., & Wang, C. (2026). Potential insurance implications of climate-smart production practices [Presentation]. Beltwide Cotton Conferences Economics and Marketing Conference, San Antonio, TX.

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Citation

Campos, P., Ahlawat, H., Pilania, K., Buffington, S., Bagavathiannan, M., Morgan, G., & Basinger, N. (2026). How cool are living mulches? Cool-season perennial grasses for weed management in cotton [Presentation]. Weed Science Society of America, Raleigh, NC.

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Citation

Campos, P., Ahlawat, H., Pilania, K., Buffington, S., Bagavathiannan, M., Morgan, G., & Basinger, N. (2026). How cool are living mulches? Cool-season perennial grasses for weed management in cotton [Presentation]. Southern Cover Crop Council Conference, Gainesville, FL.

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Citation

Campos, P., Ahlawat, H., Pilania, K., Buffington, S., Bagavathiannan, M., Morgan, G., & Basinger, N. (2026). Evaluating perennial cool-season grasses for weed suppression in cotton production [Presentation]. Southern Weed Science Society, Nashville, TN.

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Citation

Abeyasinghe, G., Wagner, T., Santos, J., Parunandi, S., Nagy, E., Mohsin, F., Bagavathiannan, M., & Antony-Babu, S. (2026, January 7–9). Organic cotton practices reshape hyphosphere microbiota and modulate the microbial response to *Fusarium oxysporum* f. sp. *vasinfectum* Race 4 [Oral presentation]. Disease Council, Beltwide Cotton Conference, San Antonio, TX.

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Citation

Parunandi, S. S., Abeysinghe, G., Bagavathiannan, M., Rajan, N., & Antony-Babu, S. (2026, January 7–9). Ecosystem resilience and microbial diversity in cotton systems under organic, transitional, and conventional management [Oral presentation]. Beltwide Cotton Conference, San Antonio, TX. Third Prize, Student Competition.

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Author ORCID(s)

Citation

Wagner, T. A., Abeyasinghe, G., Santos, J., & Antony-Babu, S. (2026, January 7–9). Functional dissection and experimental reassembly of the hyphosphere FOV4 pathobiome [Poster presentation]. Beltwide Cotton Conference, San Antonio, TX.

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Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Santos, J., Abeyasinghe, G., Wagner, T., Parunandi, S., Bagavathiannan, M., & Antony-Babu, S. (2025). A hyphal-baiting method for enriching hyphosphere bacteria associated with *Fusarium oxysporum* f. sp. *vasinfectum* Race 4 [Poster presentation]. BESC Symposium, Department of Plant Pathology and Microbiology, Texas A&M University, College Station, TX

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Author ORCID(s)

Citation

Abeyasinghe, G., & Antony-Babu, S. (2025). Hyphosphere recruitment dynamics of *Fusarium oxysporum* f. sp. *vasinfectum* Race 4 in cotton soils [Oral flash presentation]. 9th Annual Texas A&M Postdoctoral Research Symposium, Texas A&M University, College Station, TX.

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Citation

Sanyal, D., Mayo, K., Thodeti, P., Aswin, D., Stackpole, C., Fahlgren, K., & Norton, R. (2026). Soil management in the irrigated desert cotton production systems [Presentation]. Cotton Production Seminar, Casa Grande, AZ.

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Author ORCID(s)

Citation

Sanyal, D., Mayo, K., Thodeti, P., Aswin, D., Stackpole, C., Fahlgren, K., & Norton, R. (2026). Updates from regenerative cotton research in Arizona [Presentation]. Crop Production Seminar, Safford, AZ.

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Digital Object Identifier (DOI)

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Citation

Seigerman, C. K., & Thompson, J. J. (2025, November). How King Cotton continues to haunt Georgia cotton farming [Oral presentation]. American Anthropological Association Annual Meeting, New Orleans, LA.

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Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Leon, R. G., Perera, C., Moore, V., Reberg-Horton, S. C., Mirsky, S. B., & Kissing Kucek, L. (2025). Cereal rye (*Secale cereale*) cultivars differ in allelopathic activity, inhibiting growth in unique ways depending on weed and crop species [Presentation]. CANVAS 2025, Salt Lake City, UT.

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Digital Object Identifier (DOI)

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Citation

Goldsmith, A., Almeida, E., Dobbs, A., & Leon, R. (2025). Evaluation of Brassica carinata A. Braun crop safety to potential herbicide programs [Presentation]. Weed Science Society of America and Canadian Weed Science Society Annual Meeting, Vancouver, Canada.

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NIFA Support Acknowledged

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Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Goldsmith, A., Dobbs, A., & Leon, R. (2025). Comparison of cover crop biomass predictions to summer annual weed escapes using spatial analysis techniques [Presentation]. Weed Science Society of America and Canadian Weed Science Society Annual Meeting, Vancouver, Canada.

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Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Kariyawasam Hetti Gamage, L. R., Poudyal, C., Bagavathiannan, M., & Rajan, N. (2025). Evaluating cover crop termination strategies for sustainable cotton production in raised bed and flat planting systems [Abstract]. ASA–CSSA–SSSA International Annual Meeting, CANVAS 2025, Salt Lake City, UT.

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Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Gamage, L. K. H., Poudyal, C., & Rajan, N. (2026). Assessing the impact of raised bed and non-raised bed planting systems with cover crops on cotton productivity in the Texas Blackland region [Proceeding paper]. Beltwide Cotton Conferences, San Antonio, TX.

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Citation

Gamage, L. K. H., Poudyal, C., & Rajan, N. (2026). Assessing the relationship between UAV-derived NDVI and nitrogen use dynamics in cotton under variable cover crop and bed management systems [Proceeding paper]. Beltwide Cotton Conferences, San Antonio, TX.

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Citation

Gamage, L. K. H., Poudyal, C., & Rajan, N. (2026). Assessing the impact of raised bed and non-raised bed planting systems with cover crops on cotton productivity in the Texas Blackland region [Oral presentation]. Beltwide Cotton Conferences, San Antonio, TX.

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Citation

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Digital Object Identifier (DOI)

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Citation

Singletary, M. M., Dotray, P. A., Burke, J., Keeling, W., Breaux, Y., Jones, S., & Bagavathiannan, M. V. (2026). Cereal rye seeding rate and termination method on weed suppression in organic cotton. In Proceedings of the Southern Weed Science Society Annual Meeting, Nashville, TN.

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Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Singletary, M. M., Dotray, P. A., Breaux, Y., Jones, S., & Bagavathiannan, M. V. (2026). Organic herbicides for terminating a cereal rye cover crop. In Proceedings of the Southern Weed Science Society Annual Meeting, Nashville, TN.

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2025

NIFA Support Acknowledged

YES

Digital Object Identifier (DOI)

Author ORCID(s)

Citation

Loura, D., & Bagavathiannan, M. (2025). A Multi-State Precision Regenerative Cotton Research Network across the US Cotton Belt [Poster presentation]. CANVAS 2025.

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Author ORCID(s)

Citation

Loura, D., Rajan, N., Poudyal, C., Goldsmith, A., Leon, R. G., Gholson, D. M., & Hamberg, R. (2025). Cover Crop Establishment Methods on Raised Beds: Impact on Weed Dynamics [Poster presentation]. CANVAS 2025.

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Author ORCID(s)

Citation

Loura, D., Rajan, N., Poudyal, C., Goldsmith, A., Leon, R. G., Gholson, D. M., & Hamberg, R. (2026). Weed Suppression Dynamics under Different Cover Crop Planting Methods on Beds [Poster presentation]. Beltwide Cotton Conference, 2026.

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Digital Object Identifier (DOI)

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Citation

Loura, D., & Bagavathiannan, M. (2026). A Multi-State Precision Regenerative Cotton Research Network across the US Cotton Belt [Poster presentation]. Beltwide Cotton Conference, 2026.

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Digital Object Identifier (DOI)

10.3389/fmicb.2025.1661556

Author ORCID(s)

Citation

Antony-Babu, S., Abeysinghe, G., Thomas, V. E., Hockenbury, C., Parunandi, S. S., Dasgupta, A., & Pierson, E. A. (2025). Integrated metabarcoding and culture-dependent assessments reveal *Pseudomonas* as dominant hyphosphere-pathobiont in Race 4 *Fusarium* wilt pathogen of cotton. *Frontiers in Microbiology*, 16, 1661556.

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Digital Object Identifier (DOI)

10.1017/wsc.2025.10062

Author ORCID(s)

Citation

Jackson, A. W., Moore, V. M., Reberg-Horton, C., Mirsky, S. B., & Leon, R. G. (2025). Seed age changes the germination response of weed species to cereal rye (*Secale cereale*) allelopathy. *Weed Science*, 73(e9), 1–6.

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YES

Digital Object Identifier (DOI)

10.1017/wsc.2025.3

Author ORCID(s)

Citation

Goldsmith, A., Austin, R., Cahoon, C. W., & Leon, R. G. (2025). Predicting maize yield loss with crop-weed leaf cover ratios determined with UAS imagery. *Weed Science*, 73(e22), 1–9.

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Digital Object Identifier (DOI)

10.1002/cft2.70022

Author ORCID(s)

Citation

Oreja, F. H., Mahoney, D. J., Jordan, D. L., Jennings, K. M., Vann, M., & Leon, R. G. (2025). Crop rotation and herbicide program effects on Palmer amaranth and common ragweed population growth rate. Crop, Forage & Turfgrass Management, 11, e70022.

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Other Products

{Nothing to report}

Changes/Problems

Overall, the project objectives and major work plans remained on track during the reporting period. No major changes to the overall project goals, objectives, or approach are being requested at this time. Collaborators encountered some practical field, laboratory, staffing, equipment, and coordination-related challenges that are common in multi-location agricultural research projects. These challenges were addressed through improved planning, adjusted field operation timing, additional coordination with growers and partners, prioritization of laboratory activities, and strengthened staffing support.

Objectives 1.1, 1.2, and 1.3: Some delays were experienced in completing specific laboratory activities because essential equipment, laboratory supplies, and shared facilities were in high demand due to multiple students and projects using the same resources. The team continued to coordinate facility use, prioritize sample processing and analyses, and adjust timelines to keep the work moving forward. In a few research locations, on-farm soil sampling schedules were slightly shifted due to coordination and communication gaps. These issues have been addressed, and the team is back on track, with sampling activities scheduled for Spring 2026.

Several team members also experienced field sampling delays due to dry soil conditions, weather variability, grower field operations, and limited personnel availability during certain periods because of class schedules and other commitments. These challenges were managed by improving communication with growers, coordinating field visits earlier, and hiring additional help for specific sampling dates. For the living mulch component, late crop harvest affected grass planting timing, and herbicide applications were delayed in some cases. The team plans to improve coordination and timing for these activities in the next field season.

Work is ongoing to expand the digital solution to broader stakeholder groups. This effort requires additional steps to adapt, refine, and scale the tool for wider use across cotton production systems. The team continues to work on improving the functionality and applicability of the tool for different users and production conditions. Minor field-related challenges were experienced at some locations, including weather variability and timing constraints during field operations. These occasionally affected the scheduling of treatment applications and sampling. The team managed these issues by adjusting field operation timing while maintaining the experimental protocol and completing planned measurements as consistently as possible.

Objective 2: There was an initial delay in recruiting a student for the irrigation scheduling study. The student is now in place and actively working on the project, and progress is expected to continue in this area. Another challenge was limited cottonseed availability during the previous season, which allowed only limited replications and reduced the ability to detect stronger cultivar differences. To improve the robustness of future results, the team harvested additional seed and plans to establish five replications at both the Alabama and Texas locations in the 2026 field season.

Team members also experienced a delayed cotton planting window in 2025 due to unusually wet weather and a disk harrow malfunction during a critical period. Although cotton was planted later than ideal for maximum production, the experiment remained valuable because the primary focus was non-chemical, reduced-tillage weed management. In addition, software-related timing issues with the autonomous tractor affected timely weed control treatments. These experiences provided practical learning opportunities and will help improve future implementation of autonomous weed management technologies.

Objective 4: The social science team experienced a delay in recruiting participants for the April 2026 focus groups. This was mainly because additional time was needed to build stronger working relationships with Extension officers and regional partners. The team addressed this challenge by working more closely with the Soil Health Institute to connect with cotton farmers in the region and improve participant recruitment.

No special or additional reporting requirements beyond those specified in the award terms and conditions are being requested at this time.